

Health measures for the prevention project

Construction, validation, and the choice among them

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Built and validated by `data_cleaning/03--06` and `descriptives/01--07`; every number here is printed by those scripts' self-validation output. This note records which UKHLS health variables the project uses, how each measure is constructed and why, what the new graded-response measures deliver against everything that came before them (the original physical GRM, PCS/MCS, the physical-only composite, SF-6D, GHQ), and how the ever-diagnosis, dimensionality and ceiling questions resolve.

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1 The variable inventory

Measurement models are restricted to items asked in (nearly) every wave, so a score means the same thing wherever it lands in the panel. Presence was verified against the raw headers of all fifteen `indresp` files, labels against the UKDA-6614 dictionaries.

Variables	Waves	Used
SF-12, all 12 items (self-completion preferred)	all 15	both GRMs; PCS/MCS re-build; SF-6D
GHQ-12 items (<code>scghqa..1</code> , 4 categories)	all 15	mental GRM
Impairment list <code>disdif1..12</code> , gated on <code>health</code>	all 15	physical GRM (FUNC testlet)
Long-standing illness (<code>health</code>)	all 15	gate for <code>disdif</code> ; descriptive
Diagnosed conditions (<code>hcond*</code> families, codes 1–17)	all 15 (§2)	condition items; sensitivities
Health satisfaction (<code>sc1fsat1</code>)	all 15	validation outcome only
SWEMWBS; ADL module; loneliness	1/4/7/10/13; 7/9/11/13; 9–15	excluded: fail the every-wave rule
Hospital out- and in-patient attendance (<code>h12gp</code> , <code>hosp</code> ; §8 on the absent GP count)	7–15	criterion-validity outcomes
Height/weight (self-report)	wave 1 only	excluded (biomarkers are UKDA-7251, not held)

2 Chronic conditions

2.1 Construction

UKHLS asks the full “ever diagnosed” inventory (`hcond1..17`) at a person’s *first* interview; later waves record conditions newly reported since the last interview under a name that changes with questionnaire redesigns — `hcondn{i}` in waves 2–9; from the wave-10 redesign, `hcondcode{i}` and

`hcondncode{i}`, both present in waves 10–15 and both used — all dictionary-verified. The running indicator (*ever had condition i by wave w* $\Leftrightarrow$ *reported in any family at any wave $\leq w$*) is monotone and defined from the first inventory onward (76% of person-waves). Only the wave-1 code list is used, so the definition is identical in every wave. Per-condition indicators, diagnosis ages and group counts live in `chronic_conditions.parquet`; nothing is collapsed before modelling.

This corrects two defects in the earlier EIT count, which used `hcondns*` (“still has *newly diagnosed* condition”) as its new-report family: that family has no codes 6 and 7 at all — every in-panel heart attack and stroke was missed — and it drops new diagnoses reported as already resolved. Our count is $\geq$ the EIT count on 100.000% of 458k person-waves (exact on 94.2%; 26,516 documented additions, concentrated in MI, stroke and cancer).

2.2 Why “ever”, not “current”

The “still has” family (`hconds##`) is asked *only at the wave a condition is reported*: at wave 6, of 5,090 ever-asthma respondents, $\sim$ 570 were routed to it. A per-wave “current” variable would be $\sim$ 90% carry-forward imputation. Beyond measurability, self-reported remission conflates control with cure — managed hypertension and diabetes read as “no longer have”, the mechanism behind the under-reporting found against measured disease (Johnston, Propper & Shields 2009) and GP records (Kriegsman et al. 1996: good agreement for diabetes, poor for heart disease and arthritis). The monotone ever-indicator matches the deficit-accumulation tradition (Mitnitski, Mogilner & Rockwood 2001; Searle et al. 2008). Its genuine cost — old diagnoses shadowing current health — is quantified in §5, not assumed away.

2.3 Why grouped, not seventeen indicators

Grouping is the testlet logic applied to diagnoses. CHD, angina, MI, CHF and stroke are one disease process reported several ways; as separate items they would violate local independence exactly as the `sf2a/sf2b` pair did before the original GRM collapsed them (Wainer & Kiely 1987; Yen 1993). Rare conditions (1–2% prevalence) also make weak, unstable IRT items — minimum prevalence is an explicit deficit-inclusion criterion in Searle et al. (2008) — and groups keep a stable clinical meaning while the code list grows (cf. Barnett et al. 2012). An ordinal within-group count (0/1/2+) keeps a severity gradient the way `PF = sf2a + sf2b` does. The groupings follow the disease process rather than the questionnaire. **CVD** collects the several names UKHLS gives to one underlying process — atherosclerosis and its consequences — which is why a respondent can plausibly report three of its five items at once. **METAB** keeps the two long-term managed conditions apart from those acute events: hypertension and diabetes are treated rather than survived, and both are far commoner. **RESP** spans the chronic airway diseases; **MSK** is the single commonest cause of physical limitation in the data; **CANCER** stays alone because it is a distinct process and the one most relevant to mortality. **OTHER** pools the remainder, none of which reaches 5%, so that no item has to be estimated on a prevalence near 1%.

The seventeen conditions, their groups, and what each one is:

Group	Condition	What it is	Ever
CVD	angina	chest pain when the heart's blood supply cannot meet demand, typically on exertion	2.9%
	heart attack	a blocked artery kills part of the heart muscle	2.7%
	coronary heart disease	the arteries feeding the heart narrow with fatty deposits	2.4%
	stroke	blood supply to part of the brain is cut off, damaging brain tissue	2.1%
	congestive heart failure	the heart pumps too weakly, causing breathlessness and fluid retention	0.8%
	<i>any of the above</i>		7.4%
METAB	high blood pressure	blood pressure persistently above the healthy range	21.4%
	diabetes	the body cannot regulate blood sugar properly	7.2%
	<i>either</i>		24.7%
RESP	asthma	airways narrow and inflame in episodes, causing wheeze and breathlessness	15.4%
	chronic bronchitis	long-term airway inflammation with persistent cough and phlegm	2.0%
	emphysema	the lungs' air sacs are permanently damaged, so breathing is laboured	0.8%
	<i>any of the above</i>		16.7%
MSK	arthritis	painful inflammation and stiffness of the joints	16.2%
CANCER	cancer or malignancy	any malignant tumour	5.1%
OTHER	hypothyroidism	an under-active thyroid gland, slowing the body's metabolism	4.2%
	liver condition	any long-term liver disease, such as cirrhosis or hepatitis	2.2%
	hyperthyroidism	an over-active thyroid gland, speeding the metabolism up	1.3%
	epilepsy	a tendency to recurrent seizures from abnormal electrical activity in the brain	1.3%
	<i>any of the above</i>		8.1%
DEPR[†]	clinical depression	a diagnosed depressive disorder, as distinct from transient low mood	8.2%

“Ever” is the share of person-waves at which the condition has been reported at or before that wave, among the 458,468 person-waves with an inventory. Conditions are listed within group by prevalence. [†]DEPR is the one group assigned to the mental bank rather than the physical one.

2.4 Diagnosis timing

Every inventory report carries `hconda##` (age told); every in-panel report is dated by its wave: **100.0% of ever-flags carry a usable diagnosis age**, so recency-restricted variants are exact. The stock is old: median time since diagnosis 11 years ($p_{25} = 5$, $p_{75} = 21$); at ages 65–80, 54% of flags are >10 years old, 26% >20.

3 The graded-response measures

3.1 Why separate physical and mental banks

The measurement model must stay a static map from responses to a score (dynamics are estimated afterwards), and PROMIS practice is separate unidimensional banks per domain (Cella et al. 2010). Two clean banks also make the physical–mental correlation an *estimate* rather than an artifact: the SF-12’s orthogonal scoring forces $r(\text{PCS}, \text{MCS}) = 0$; our corrected promax on the subscales gives 0.59; published correlated-scoring estimates are 0.62 (Farivar, Cunningham & Hays 2007) and 0.71–0.73 (Tucker et al.); the two GRM thetas correlate **0.457**. Cross-loading SF-12 items are assigned by this repo’s own UK wave-1 varimax loadings (phys/ment): PF .86/.14, RP .85/.25, BP .75/.17, GH .72/.27 → physical; MH .09/.89, RE .26/.82, SF .48/.64 → mental. **VT (energy, .55/.44) enters neither bank** (PROMIS likewise treats fatigue as its own domain); cognitive and sensory disdif items are separate WHODAS-type constructs and are excluded from both.

3.2 The banks

Every item is coded so category 1 is worst health.

Spec	Items
P-FUNC	GH, PF, RP, BP (the validated original testlets) + FUNC: count of physical disdif limitations {mobility, lifting, dexterity, coordination, personal care}, 0/1/2/3+; health =“no” is a valid structural zero (module skipped by design)
P-FULL	P-FUNC + six ever-diagnosis group items (CVD/METAB/RESP 0/1/2+; MSK/CANCER/OTHER binary)
P-REC	as P-FULL, counting only diagnoses within the last 10 years
P-TIMED	as P-FULL with each group split into recent ($\leq 10\text{y}$) and stale ($>10\text{y}$) items — the model <i>estimates</i> the recency weighting (§5)
MENT	MH ((6 – sf6a) + sf6c – 1, 9 levels), RE, SF, GHQ-positive and GHQ-negative wording testlets (19 levels each), ever-depression
COMBINED	P-FULL $\cup$ MENT: all seventeen good-coverage items in one bank (§6)

GHQ-12 is validated internationally (Goldberg et al. 1997) but its factor-analytic “dimensions” largely reflect negative-wording method effects (Hankins 2008, contra Graetz 1991). The pre-specified rule — fit item-level; collapse by wording if Yen’s Q_3 shows positive within-wording dependence — *triggered* (within-wording Q_3 max +0.26), the same cure the original GRM applied to the PF/RP pairs.

3.3 Estimation and diagnostics

Same machinery as the exactly-replicated original (Samejima 1969; marginal ML by EM over response patterns, 121-point quadrature, pooled latent $N(0, 1)$; multigroup by single year of age 20–90 for μ_a, σ_a). Samples: P-FUNC 497,423 person-years; P-FULL/P-REC/P-TIMED 387,599 (condition items exist from first inventory); MENT 379,462; COMBINED 376,019.

	P-FUNC	P-FULL	P-REC	MENT	COMBINED
max positive Q_3	-0.048	+0.144	+0.080	+0.153	+0.582 [†]
EAP identity (var + $E[\text{psd}^2]$)	1.000	1.000	1.000	1.000	0.999
r with original GRM θ	0.994	0.985	—	—	0.917

[†]the mental-pair clustering that is itself the two-dimensionality evidence; the physical side stays clean (§6).

Discriminations: the functioning items dominate — FUNC $a = 3.3$ (P-FUNC: GH 2.0, PF 3.7, RP 3.4, BP 2.4) — while the condition-group items are weak (P-FULL: MSK 1.30, CVD 1.23, METAB 0.86, OTHER 0.65, CANCER 0.60, RESP 0.41): once functioning is measured well, a diagnosis adds comparatively little about *current* physical health — with arthritis, a condition that *is* impaired functioning, the strongest of them. On the mental side, the GHQ negative-wording testlet ($a = 3.15$) and MH (2.75) carry the bank; the ever-depression diagnosis is again weak (0.97).

3.4 Routing: who is asked what

Two items in the physical bank are asked conditionally, and both deserve stating plainly.

The functioning testlet is gated on long-standing illness. The `disdif` impairment list is only put to respondents who report a long-standing illness, so FUNC takes its best category structurally for the 66.2% of scored rows with `health == 2`. Only 33.8% of rows actually answer the module, and among those FUNC is well spread (17/14/18/51% across its four levels). The item is therefore 83.4% top category overall: it says nothing about differences among the healthy and everything about gradations among the ill — which is exactly what its estimated hurdles (-1.83 , -1.42 , -1.05) and its high discrimination ($a = 3.4$) encode. This is a feature for our purposes, because the ill tail is where the other items have run out (§7), but it does mean FUNC is partly a re-reading of the long-standing-illness question. It is not double counting in a way the model minds: FUNC’s largest residual correlation with any other item is $+0.031$. A further 6,451 rows report an illness but leave the module unanswered; those are treated as missing, not as zero limitations.

The diagnosis items exclude the BHPS sample entirely. The condition inventory is asked at a person’s first interview, and wave 2 — where the BHPS continuers joined UKHLS — has no inventory at all. They are then never routed back to it: only 4.4% of the 16,204 people first interviewed at wave 2 ever receive one, against 94.8% of wave-1 entrants and 70.7% of wave-3 entrants. P-FULL therefore drops 14,721 people outright (17.4% of persons, 109,470 person-waves), and this is a survey-design exclusion rather than attrition.

Whether that matters depends on the use. On health it looks benign: the dropped people are indistinguishable from the kept on every measured dimension (mean GH 3.32 vs 3.32, PF 4.23 vs 4.24, FUNC 3.66 vs 3.67). On *panel structure* it is not benign at all: the excluded BHPS members are the longest-observed people in the study (7.4 observations each against 5.5 for those kept). Any diagnosis-bearing measure therefore discards precisely the respondents with the richest longitudinal information, which is a poor trade for trajectory work (§10).

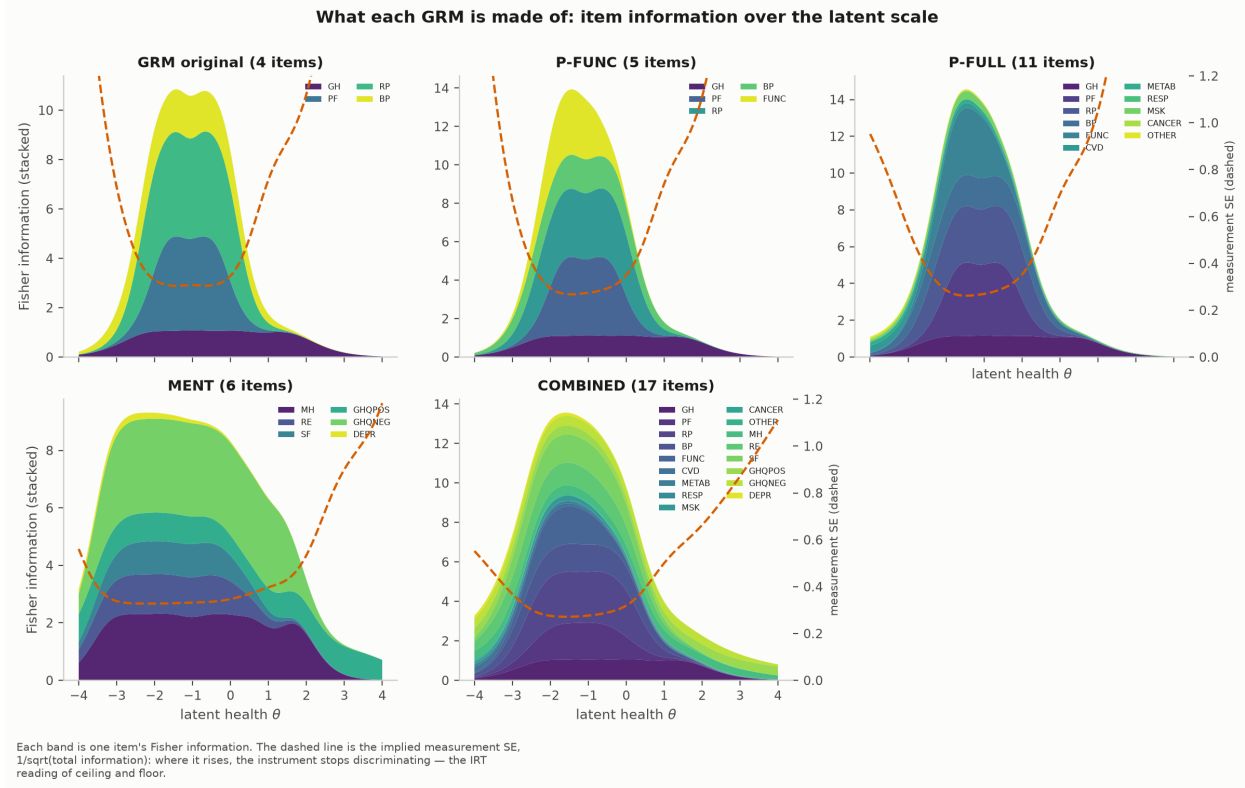


Figure 1: What each GRM is made of: stacked item information over the latent scale, with the implied measurement SE (dashed, right axis); each band is one item, ordered as in the legend. **Panels:** **GRM original** is the four-testlet physical measure replicated from the EIT note; **P-FUNC** adds the functioning testlet; **P-FULL** adds the six diagnosis groups; **MENT** is the mental bank; **COMBINED** is all seventeen items at once. *Physical items, from the SF-12:* **GH** general health (the single self-rated-health question), **PF** physical functioning (moderate activities, climbing stairs), **RP** role-physical (cutting down or being limited at work because of physical health), **BP** bodily pain; **FUNC** is the count of Equality Act physical limitations (mobility, lifting, dexterity, coordination, personal care). *Diagnosis groups:* **CVD** cardiovascular events, **METAB** diabetes and high blood pressure, **RESP** chronic airway disease, **MSK** arthritis, **CANCER**, **OTHER** the pooled remainder of thyroid, liver and epilepsy (§2.3). *Mental items:* **MH** mental health (calm; downhearted), **RE** role-emotional (cutting down or working less carefully because of emotional health), **SF** social functioning (health interfering with social activities), **GHQPOS** and **GHQNEG** the positively- and negatively-worded halves of the GHQ-12, **DEPR** ever-diagnosed clinical depression. **FUNC** adds a broad band of precision on the ill side; the condition items are visibly thin; GHQ-negative dominates the mental bank; the combined bank is tallest and widest but its SE still rises sharply above $\theta \approx +1$.

4 How the measures relate

Correlations over all person-years (descriptives/01):

	θ phys (FULL)	θ ment	θ combined
original GRM (4 testlets)	0.985	0.48	0.92
UKHLS PCS	0.904	0.23	0.76
physical-only composite	0.946	0.48	0.90
UKHLS MCS	0.27	0.882	0.58
GHQ-12 Likert	-0.38	-0.904	-0.64
SF-6D utility	0.71	0.799	0.88
chronic count	-0.58	-0.21	-0.51
θ phys $\times$ θ ment		0.457	

Three reads. The physical GRM is the object the project already validated, with functioning attached. The mental GRM is the MCS/GHQ construct without the orthogonality artifact. SF-6D correlates *more* with the mental theta (0.80) than the physical one (0.71) — and its closest cousin is the *combined* bank (0.88): the SF-6D was a mixed measure all along.

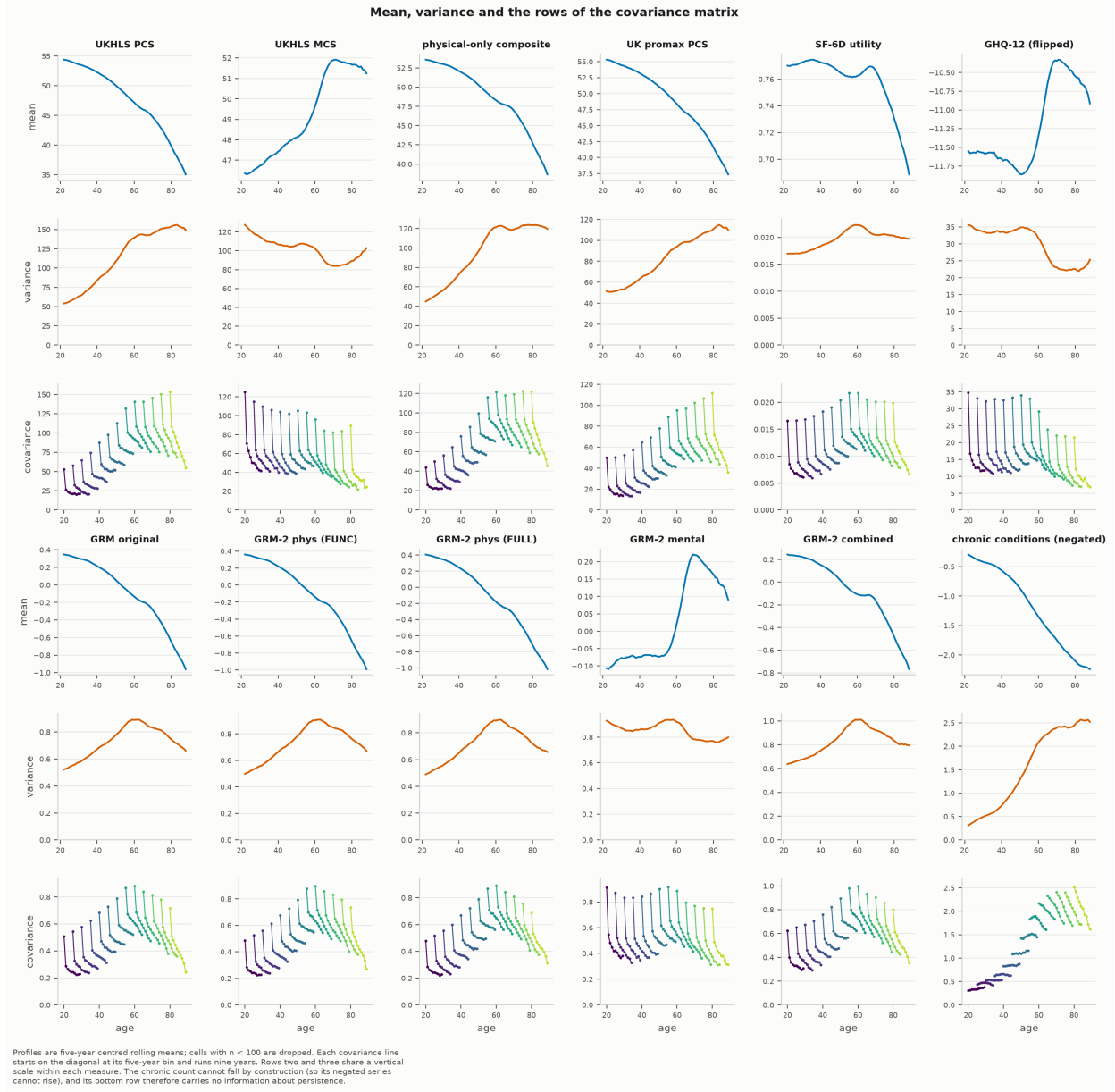


Figure 2: Mean, variance and the rows of the covariance matrix (the empirics note’s anatomy) for the full metric suite, two banks of six. GHQ and the chronic count are *negated*, so on every panel higher means better health. The mental family (MCS, GHQ, mental GRM) shows the textbook late-life wellbeing improvement; the physical measures share one anatomy — accelerating mean decline, rising-then-plateauing dispersion, fanning covariance rows; the chronic count gives the physical mean and variance reads from items fully disjoint from the SF-12, while its covariance row carries no persistence information because the count cannot fall (so its negated series cannot rise).

Two cautions on the mental hump, because on an auto-scaled axis it looks dramatic. It is modest in size: mean GHQ falls from 11.75 at age 55 to 10.24 at 70, which on a 0–36 scale with a standard deviation of 5.62 is 0.27 sd; the axis spans less than two points, which is what makes it read as a step change. And *most of it is composition rather than ageing*: among the 1,461 people observed in

both windows, mean GHQ improves by only 0.44 points between ages 53–57 and 68–72, against a cross-sectional gap of 1.42 over the same ages, so under a third of the apparent improvement survives holding the people fixed. The remainder is selective non-response and attrition — GHQ completion falls from 91% at age 70 to 77% at 90 — together with cohort differences. This applies to every panel of the figure: these are pooled cross-sections in which age, cohort and period are confounded, and they are descriptive scaffolding rather than within-person trajectories. The mixture models of §9 do use within-person information and are not subject to it.

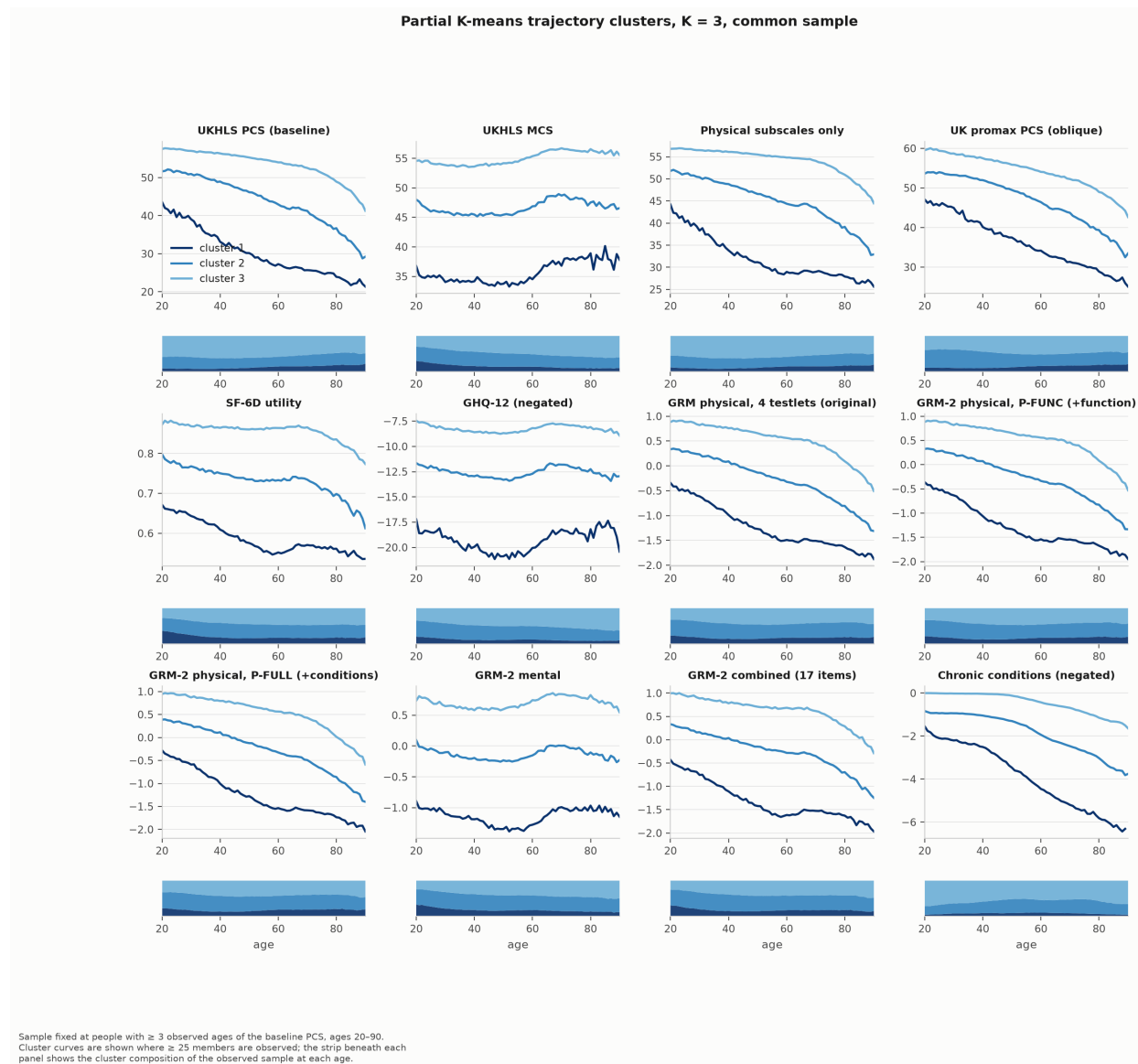


Figure 3: Partial K-means trajectory clusters ($K = 3$, 50 seeded inits, sample fixed at people with ≥ 3 observed baseline-PCS ages, ages 20–90), across the same twelve measures as Figure 2. GHQ and the chronic count are negated so every panel runs higher = better health. The chronic count is monotone by construction, so its three bands can only fall and never cross — a property of the measure, not a finding about people.

Partial K-means (Figure 3): changing the *scoring* moves the typology modestly while changing the *construct* redraws it. Against the PCS baseline, on the common sample:

Measure	ARI	% same cluster	chance
Physical subscales only	0.667	87.9	43.0
UK promax PCS (oblique)	0.464	79.4	39.5
P-FUNC	0.442	76.9	38.1
GRM original (4 testlets)	0.425	75.6	37.7
P-FULL	0.409	74.9	37.5
GRM combined (17 items)	0.266	66.1	36.8
SF-6D utility	0.185	60.0	37.4
Chronic conditions (negated)	0.120	56.1	44.3
GHQ-12 (negated)	0.079	51.4	40.8
GRM mental	0.074	49.2	37.1
UKHLS MCS	0.053	48.0	40.0

The physical measures agree substantially with the baseline whatever the scoring — 0.41 to 0.67 — and the ordering is instructive: the composite that merely re-weights the same subscales agrees most, the IRT re-spacings sit in the middle, and the further a measure moves from the PCS’s item set the lower it falls. The mental measures sit at the bottom, essentially at chance: MCS at ARI 0.053 and the mental GRM at 0.074, with same-cluster rates of 48% and 49% against chance rates of 40% and 37%. The combined bank lands between the two families at 0.27, which is what a physically-dominated mixture of them should do. So the physical typology is robust to how physical health is scored, and the mental dimension partitions people almost independently of it — the same conclusion the two-bank correlation of 0.457 reaches by a different route.

5 The ever-diagnosis question, resolved

The misgiving: historical diagnoses accumulate mechanically with age, so a diagnosis-bearing measure could manufacture “decline” out of stale paperwork. Four reads.

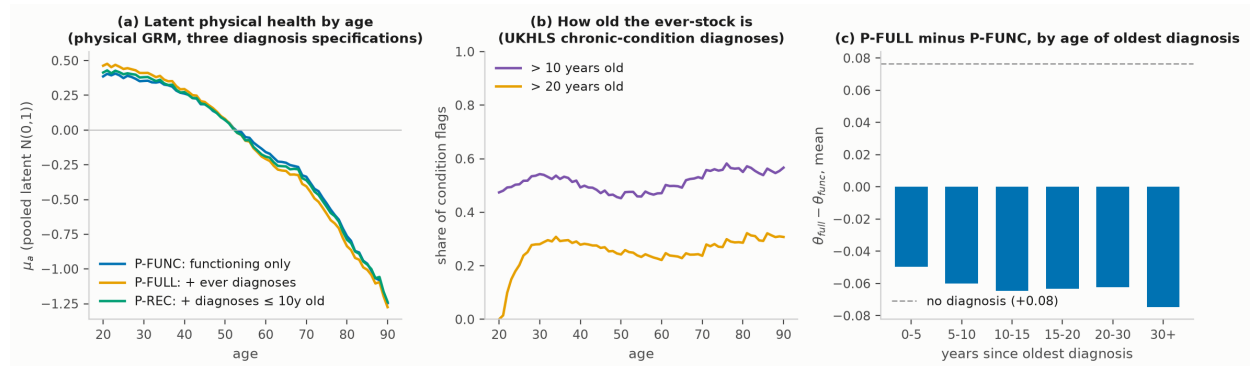


Figure 4: The ever-diagnosis question, all three panels on the physical GRM. (a) latent age profiles μ_a under P-FUNC (functioning only), P-FULL (plus ever-diagnoses) and P-REC (plus diagnoses at most ten years old); (b) the staleness of the UKHLS chronic-condition stock; (c) the per-person gap $\theta_{P-FULL} - \theta_{P-FUNC}$ against the age of that person’s oldest diagnosis.

- (1) **The aggregate effect is real but bounded.** $\mu(80) - \mu(30)$ is -1.117 SD functioning-only, -1.247 with ever-diagnoses, -1.170 with only diagnoses ≤ 10 years old. Diagnoses steepen the measured lifecycle decline by 0.130 SD (12%), and 59% of that extra steepness comes from diagnoses more than a decade old — leaving the stale-accumulation component at $\sim 6\%$ of the total decline.
- (2) **The potential was much larger than the realised effect.** Half the ever-stock is >10 years old at every age past 40 (§2.4). The damage stays small because the estimator protects itself: items that cannot predict the functioning responses get low discriminations, so stale-heavy condition items simply carry little weight.
- (3) **The items cannot tell stale from fresh — which cuts both ways.** The person-level penalty $\theta_{\text{full}} - \theta_{\text{func}}$ is flat in the age of the oldest diagnosis (-0.05 SD at 0–5 years, -0.075 at 30+). Read as measurement of *current* health, that is exactly the mechanism feared: no within-item discounting exists. It is an argument *against trusting ever-items as current-state measures*, and simultaneously the reason their damage is capped.
- (4) **Asked directly, the data mostly decline to discount.** P-TIMED splits every group into recent (≤ 10 y) and stale (>10 y) items and lets the model estimate their separate discriminations — the in-framework analogue of an estimated recency-weight function:

Group	a recent	a stale	ratio
CVD	1.19	1.21	0.99
METAB	0.59	0.81	0.72
RESP	0.68	0.29	2.31
MSK	0.83	1.22	0.68
CANCER	0.59	0.56	1.05
OTHER	0.63	0.63	1.00

Only respiratory disease wants a discount (remitting childhood asthma). For arthritis and the metabolic conditions the *stale* item is more informative — progressive diseases, where an old diagnosis means longer duration and worse current health. The mechanistic-accumulation story is really a story about *remitting* conditions; for progressive conditions the old diagnosis is legitimately informative.

Related literature. Estimating recency weights on past exposures is the weighted-cumulative-exposure literature (Sylvestre & Abrahamowicz 2009; review in Kelly et al. 2024; R package WCE), with spline-estimated weight-by-recency functions — P-TIMED is its two-knot IRT analogue. Letting item parameters vary continuously with a covariate such as time since diagnosis is moderated nonlinear factor analysis (Bauer & Hussong 2009; regularised variants for DIF detection), the natural next refinement if a smooth weighting is ever needed. The simpler administrative-data practice is a fixed lookback window (Preen et al. 2006), which P-REC implements. Health-state valuation handles the same issue with acute vs. post-acute utilities for events like stroke.

Verdict. For lifecycle *dynamics*, use P-FUNC (or the original GRM, $r = 0.994$): a pure current-state instrument. P-FULL is a sound descriptive companion whose accumulation bias is now characterised and bounded (~ 0.08 – 0.13 SD over the lifecycle, concentrated in remitting conditions); P-REC and P-TIMED are its sensitivity band. Where conditions matter substantively, model them as separate channels in the Bayesian mixture, where “ever” is the correct state variable.

6 One dimension or two? The combined GRM

The combined bank (all seventeen items, one latent dimension) fits, scores, and correlates sensibly ($r = 0.92$ with the physical bank, 0.75 with the mental, 0.88 with SF-6D). But it fails the assumption the others satisfy: after conditioning on its single dimension, the mental items remain strongly positively correlated — $Q_3(\text{MH}, \text{GHQ}^-) = +0.58$, $Q_3(\text{GHQ}^+, \text{GHQ}^-) = +0.54$, $Q_3(\text{MH}, \text{RE}) = +0.33$ — while the physical side stays clean (max phys-phys $Q_3 +0.20$). That residual clustering is the data saying the mental items share a dimension the combined score cannot represent: with $r(\theta_P, \theta_M) = 0.457$, one dimension is a genuine misspecification, not a summary. The same criteria table shows the cost in substance: the combined score reintroduces a mental-health artifact of -0.82 SD at a fixed physical profile (the physical banks: 0.00), and its cohort spread collapses to 0.06 SD not because it is clean but because physical health improving and mental health deteriorating across cohorts *cancel* inside it — precisely the mixing the EIT note warned about. Use it only where a single index is externally imposed; a bifactor model (general + physical/mental group factors) would be the defensible one-number construction if that need becomes real.

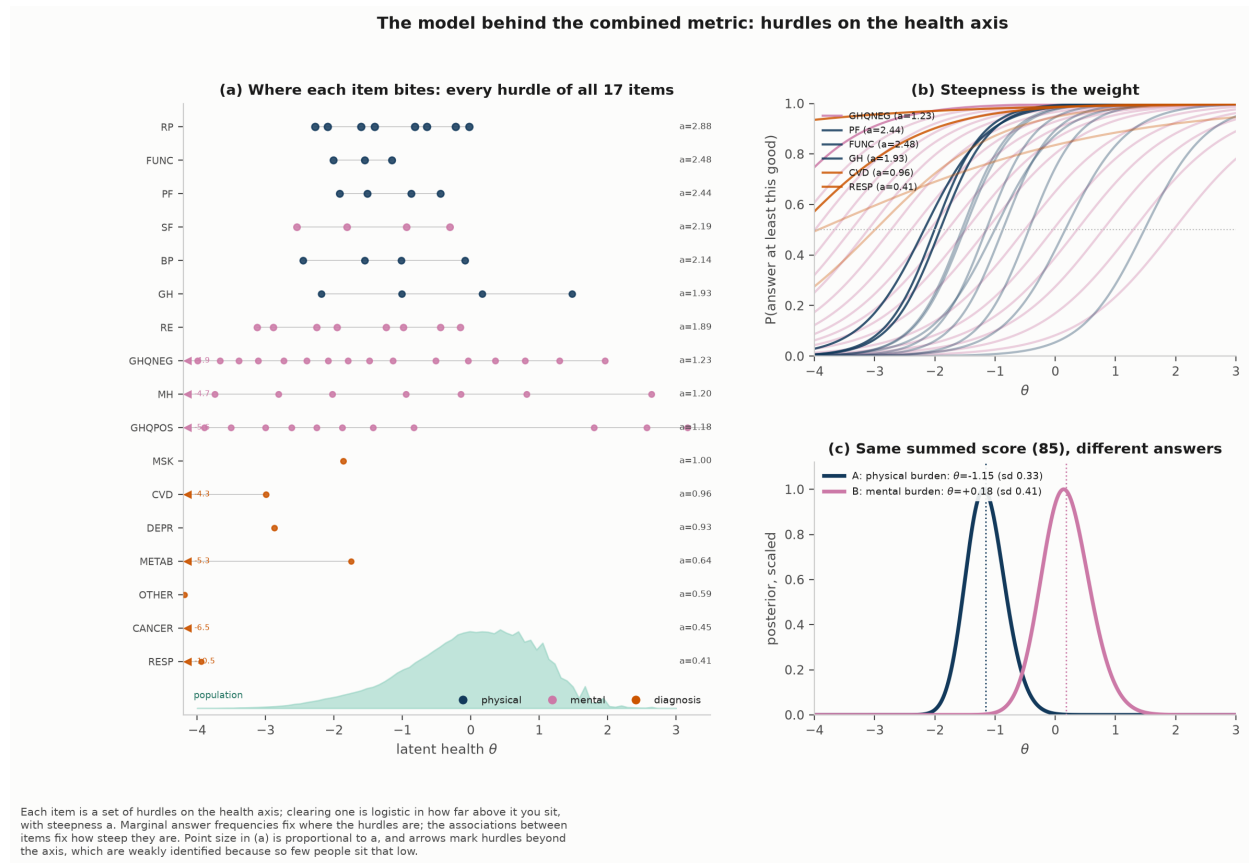


Figure 5: The model behind the combined metric, in the empirics note’s hurdles-on-the-health-axis form, for the most all-encompassing bank. (a) every hurdle of all seventeen items against the population distribution of θ ; (b) the hurdle curves for items spanning the weight range; (c) two people whose category numbers sum to the same total.

Figure 5 draws the combined bank as a set of hurdles. Panel (a) makes the division of labour

(a) Mean age path, in own-sd units relative to age 25

(b) Dispersion, relative to age 25

(c) Running out of room: ceiling above, floor below

(d) Contamination

Figure 1 consists of four panels (a, b, c, d) illustrating the performance of various GRM versions (UKHLS PCS, SF-6D, GRM original (4 testlets), GRM-2 phys, functioning, GRM-2 phys, +conditions, GRM-2 mental, and GRM-2 combined) across ages 25-80.

Panel (a) shows the mean age path in own-sd units relative to age 25. The y-axis ranges from -1.2 to 0.4. The x-axis shows age from 25 to 80. The paths generally show a downward trend, with GRM-2 mental (pink) showing the least decline and GRM-2 phys, functioning (blue) showing the most decline.

Panel (b) shows dispersion relative to age 25. The y-axis ranges from 0.9 to 1.6. The x-axis shows age from 25 to 80. The dispersion generally increases with age, with GRM-2 phys, +conditions (purple) showing the highest dispersion and GRM-2 mental (pink) showing the lowest.

Panel (c) shows the percentage of person-years at the limit (ceiling above, floor below) for the MAXIMUM (solid line) and MINIMUM (dashed line) physical profiles. The y-axis ranges from -2 to 15. The x-axis shows age from 25 to 80. The MAXIMUM paths generally show a downward trend, while the MINIMUM paths remain near zero.

Panel (d) shows contamination (MH artifact at fixed physical profile and spread across 1945-65 cohorts, ages 50-59) in sd units. The y-axis ranges from -1.5 to 0.5. The x-axis lists the GRM versions. The MH artifact (orange bars) is negative for all versions, with GRM-2 phys, +conditions showing the largest negative value. The spread across 1945-65 cohorts (grey bars) is positive for all versions, with GRM-2 phys, +conditions showing the largest positive value.

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7 Ceiling and floor

Two different questions hide under “ceiling effects”: how much probability mass sits at the best attainable response profile, and whether the instrument can *discriminate* among the healthy. More items help enormously with the first and only marginally with the second.

Measure	ceiling %	floor %	range (sd units)	psd at top	psd at bottom
UKHLS PCS (12 items)	0.89 ^a	~0	6.5	—	—
Physical-only composite	9.95	0.91	4.1	—	—
SF-6D utility	2.98	0.18	4.7	—	—
GHQ-12 Likert	0.20	0.31	6.4	—	—
GRM original (4 testlets)	9.55	0.97	4.4	0.65	0.45
P-FUNC	9.53	0.76	4.5	0.65	0.44
P-FULL	7.16	0.00	5.3	0.64	0.56
MENT	0.11	0.03	6.5	0.60	0.45
COMBINED	0.05	0.00	7.5	0.66	0.55

^ashare at the all-best 12-item profile (the exact-maximum read is degenerate for a continuous derived score). “psd” is the EAP posterior sd at the all-best / all-worst response pattern.

The optimistic half of the hypothesis is confirmed: ceiling mass collapses from 9.5% (original GRM) to 7.2% with conditions and to 0.05–0.11% for the mental and combined banks, and the floors of P-FULL and COMBINED empty entirely. Panel (c) of Figure 6 shows where it binds: the young end, where the physical banks still park 12–15% of person-years at the maximum.

The pessimistic half is also right, and Figure 1 shows why: the posterior sd at the very top is ~0.6–0.66 *for every version* — seventeen items buy no more precision among the healthy than four, because every added item has strongly negative thresholds (all these instruments measure ill-health). Extra items *spread* the top mass into a finer ranking (ties broken by GHQ and diagnoses) without adding information where the healthy live. Genuinely fixing the ceiling requires items with positive thresholds — vigorous-functioning items of the PROMIS physical-function type — which UKHLS does not carry. For the mixture models this is a manageable, known feature: the likelihood sees ordered categories, not the score, and the ceiling category is one of them.

The floor, in depth

The floor is the mirror image, and here the added items genuinely work — their negative thresholds sit exactly where the severely ill live (`descriptives/06, floor_depth.csv`):

	measurement SE at $\theta =$			min	% < -3	% at worst	distinct θ in
	-4	-3	-2	θ		pattern	bottom 1%
GRM original	2.15	0.73	0.33	-2.61	0.00	0.97	2
P-FUNC	2.12	0.67	0.30	-2.68	0.00	0.76	3
P-FULL	0.95	0.55	0.29	-3.38	0.03	0.00	575
MENT	0.56	0.34	0.33	-3.72	0.30	0.03	1,524
COMBINED	0.55	0.37	0.28	-4.43	0.24	0.00	3,708

A note on reading Figure 1: P-FULL carries visible information out to the left edge of the axis where the other physical banks have none. That is real measurement, not an artefact of where the

axis stops. Total test information at $\theta = -4$ is 1.10 for P-FULL against 0.22 for P-FUNC, and at $\theta = -6$ it is 0.37 against 0.00; the axis is not hiding a cliff, the curve simply keeps declining. The information there comes almost entirely from the diagnosis items — CVD (34% of it), METAB (18%), OTHER (10%) — whose hurdles sit at -3.6 and -4.1 , precisely in that region. It is the opposite of a floor effect: a floor effect is information *collapsing* at the bottom, which is what the functioning-only banks do. The caveat is that a few of these thresholds are weakly identified because so few people sit that low (RESP’s first hurdle is estimated at -10.5 , CANCER’s at -6.5 in the combined bank), so the far tail should not be read as precisely located — but at $\theta = -4$ the contributing hurdles are all within the range the data actually cover.

The original GRM (and P-FUNC) have a real floor problem: the instrument goes dark below $\theta \approx -2.5$ (SE 0.7 at -3 , 2.1 at -4), the scale simply ends at -2.6 , and the entire bottom percentile collapses onto two or three response patterns — the worst-off are a spike, not a distribution. The condition items repair precisely this: their thresholds (-2.6 to -4.6) are the only ones deep in the ill tail, so P-FULL keeps SE at 0.55 at $\theta = -3$, extends the scale to -3.4 , empties the worst-pattern spike entirely, and resolves the bottom percentile into 575 distinct positions. Among the P-FUNC bottom *decile* (37.6k person-years — the population a prevention analysis targets), P-FUNC distinguishes 1,252 values, P-FULL 16,522, COMBINED essentially one per person-year. The mental bank has the best floor of all (GHQ thresholds reach -3.5 ; SE 0.34 at -3).

This sharpens the ever-diagnosis trade-off of §5: the diagnosis items’ cost is a bounded accumulation bias in the age gradient; their benefit is measurement depth among the severely ill, which functioning items alone cannot provide because severe limitation saturates them. For analyses that turn on the worst-health group — describing the bottom class, targeting — P-FULL is the better physical instrument; for population dynamics, P-FUNC remains the clean choice.

8 Convexity against healthcare use

Is healthcare use convex in health — falling steeply among the ill, flat among the healthy — and does each metric’s cardinalisation carry that shape? Following the construction-note convexity exercise, each outcome is regressed on the standardised measure and its square on a common sample (waves 7–15, all fifteen measures observed; $n = 209,042$ in-patient, 208,936 out-patient), with measures oriented so higher = better: on a declining relationship, a positive quadratic = convex. One data correction first: UKHLS mainstage has *no GP-visit count*. `h12gp`, despite its name, is labelled “hosp or clinic *out-patient* last 12 months” (the sandbox’s “GP band” was a mislabel, corrected here); the only GP variable is `servuse1` (“service use: your local doctor”, yes/no, five waves), included as a supplementary outcome.

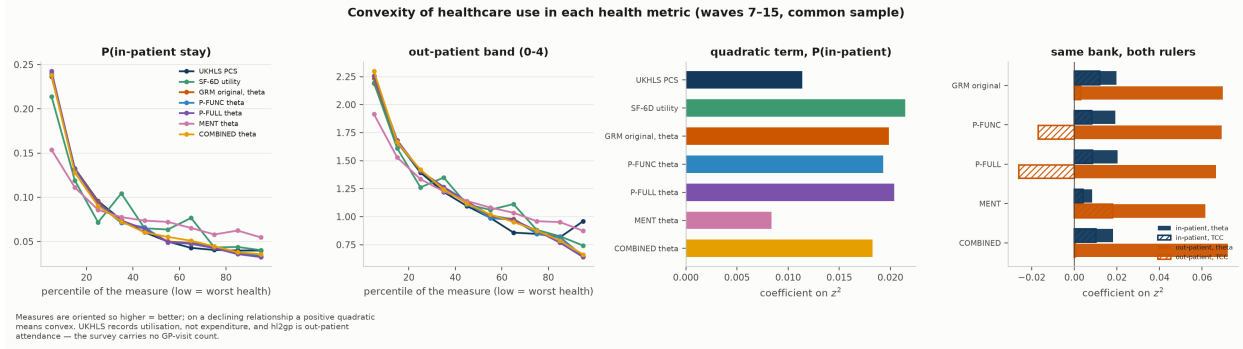


Figure 7: Healthcare use across the distribution of each measure, and the quadratic (convexity) term for in-patient admission. All fifteen measures are tested (Table above); the seven plotted in the first two panels are UKHLS PCS, SF-6D utility, the original GRM, P-FUNC, P-FULL, the mental GRM and the combined GRM, each oriented so higher means better health. The fourth panel sets each bank's θ (solid) against its TCC (hatched) for both outcomes: identical people, identical ranking, different units.

Measure	quadratic term (t)		$P(\text{in-patient})$	
	in-patient	out-patient	bottom 1%	top 1%
UKHLS PCS	+.011 (24)	+.032 (18)	.312	.042
UKHLS MCS	+.004 (8)	+.038 (22)	.128	.134
Physical-only composite	+.011 (22)	+.002 (1)	.331	.033
SF-6D utility	+.021 (40)	+.119 (61)	.289	.039
GHQ-12 (flipped)	+.003 (8)	+.007 (5)	.189	.059
<i>each bank on both rulers</i>				
GRM original, θ	+.020 (43)	+.070 (42)	.331	.033
GRM original, TCC	+.012 (23)	+.003 (2)	.331	.033
P-FUNC θ	+.019 (42)	+.069 (42)	.322	.033
P-FUNC, TCC	+.008 (17)	-.017 (-9)	.322	.033
P-FULL θ	+.020 (45)	+.067 (40)	.349	.033
P-FULL, TCC	+.009 (17)	-.026 (-15)	.349	.033
MENT θ	+.008 (19)	+.062 (37)	.206	.056
MENT, TCC	+.004 (10)	+.018 (12)	.206	.056
COMBINED θ	+.018 (44)	+.072 (49)	.331	.032
COMBINED, TCC	+.010 (24)	+.001 (0)	.331	.032

Every bank now appears on both rulers, and the last two columns are the control: they are rank statistics, so within a pair they are *identical* across θ and TCC to three decimals. Anything that moves between the two rows of a pair is cardinalisation and nothing else.

Four reads.

Utilisation is strongly convex in every latent-scale metric. The in-patient rate runs from $\sim 24\%$ in the worst decile to $\sim 3-4\%$ in the best (base rate 8.1%), with the quadratic terms at $t = 40-45$ for the GRM thetas. The gradient concentrates exactly where §7's floor analysis lives: the region in which measures differ in resolution is the region that carries the utilisation risk.

Cardinalisation, not construct, drives the convexity — and the two outcomes part

company. Each θ /TCC pair ranks people identically, so every difference below is units and nothing else.

In-patient admission is convex on both rulers. Every TCC row keeps a t between 10 and 24, roughly half the θ magnitude. Re-ruling moves the size of the effect and not the conclusion.

Out-patient attendance is where the ruler decides the answer. On θ every bank is strongly convex, $t = 37$ –49. On the TCC the same banks give $+.003$ ($t = 2$) for the original, $+.001$ ($t = 0$) for COMBINED and $+.018$ ($t = 12$) for MENT — and turn *concave* for both physical banks, $-.017$ ($t = -9$) for P-FUNC and $-.026$ ($t = -15$) for P-FULL. A t of -15 is not a null result; it is a confident statement of the opposite shape.

Convexity is therefore a property of the chosen units, and a “convex health effects” claim means nothing until the scale is fixed. Conversely **grmh** is the natural unit when something close to linear-in-utilisation spacing is wanted — the near-zero TCC quadratics are that property, stated — and θ the natural latent-normal unit for the mixture models.

The floor resolution pays off in real outcomes. P-FULL’s bottom 1% has a 34.9% admission rate against 32.2% for P-FUNC and 31.2% for the PCS: the diagnosis items isolate a genuinely sicker extreme tail, the outcome-side counterpart of §7’s 575-fold floor de-clumping.

The GP-use binary is uninformative about shape. At a 77.8% base rate ($n = 23,953$ on its five waves) it saturates: the θ quadratics are ≈ 0 for every bank. The TCC rows are uniformly negative and significantly so ($t = -5$ to -8), which is the same units effect as out-patient use rather than anything about GP visits. Bottom-1% GP use is $\sim 97\%$ against 56–74% at the top, by measure — a gradient, but the outcome’s own ceiling swamps any curvature. A true visit count would need linked administrative data.

9 Latent-class trajectories under AR(1) persistence

The measures feed the Bayesian clustering layer: $K = 3$ quadratic growth mixtures (the consolidated Stan model, partition inits with jitter, 4 chains $\times$ 1000+1000) on each channel’s frozen lifecycle contract, in three specifications — no AR term, AR(1) persistence in the class residual, and AR(1) with *within-person holdout*: each person’s last two observations are excluded from the likelihood and scored as a conditional predictive density, with the sample restricted to people observed at least five times. Class 1 is worst health throughout (the anchor convention). All eight fits converged; the batch took 15.9 h on two workers.

Channel	Spec	Persons	θ (shares)	ρ (persistence)	max $\hat{R}$	wall
PCS	AR(1)	50,194	.165 / .273 / .561	.67 / .31 / .29	1.004	8.3 h
PCS	AR(1) + holdout	39,349	.146 / .234 / .620	.67 / .26 / .30	1.006	6.2 h
GRM phys (P-FULL)	baseline	38,963	.159 / .437 / .404	—	1.003	2.0 h
GRM phys (P-FULL)	AR(1)	38,963	.328 / .360 / .312	.78 / .27 / .39	1.001	4.1 h
GRM phys (P-FULL)	AR(1) + holdout	29,901	.313 / .362 / .325	.78 / .26 / .37	1.002	2.5 h
GRM combined	baseline	38,181	.163 / .454 / .383	—	1.002	1.9 h
GRM combined	AR(1)	38,181	.290 / .379 / .331	.77 / .34 / .44	1.002	3.6 h
GRM combined	AR(1) + holdout	29,280	.271 / .377 / .352	.77 / .32 / .44	1.003	3.1 h

Persistence is strongly class-graded, on every channel. The worst-health class carries $\rho_1 = 0.67$ –0.78 against 0.26–0.44 for the other two, with posterior standard deviations of 0.003–

0.007: shocks to the least healthy class decay several times more slowly than shocks elsewhere, and the ordering is identical across three different instruments. This is the mixture-model counterpart of the EIT note’s persistent-layer anatomy, estimated jointly with the class structure rather than imposed.

The AR term, not the scale, redistributes the classes. The interim reading of these fits was that the GRM channels produce near-even class shares where the PCS stays bottom-heavy. The baselines refute the measurement-scale half of that: without an AR term, *both* GRM channels put only 16% in the worst class (.159 and .163), close to the PCS convention. Adding persistence roughly doubles that share to .29–.33. The mechanism is visible in the intercepts: the baseline worst class sits at $\alpha_1 \approx -1.6$ on all three channels, while under AR(1) the GRM worst class moves up to $\alpha_1 \approx -1.0$ and takes in many more people. Without persistence, a small extreme class must absorb the autocorrelation of repeated low observations; once ρ can do that work, the class boundary moves and a broader, less extreme worst class emerges. The PCS resists this (θ_1 .165 under AR(1)), which is consistent with its coarser floor (§7): it cannot separate the severely ill finely enough for the boundary to move.

Class trajectories and shares

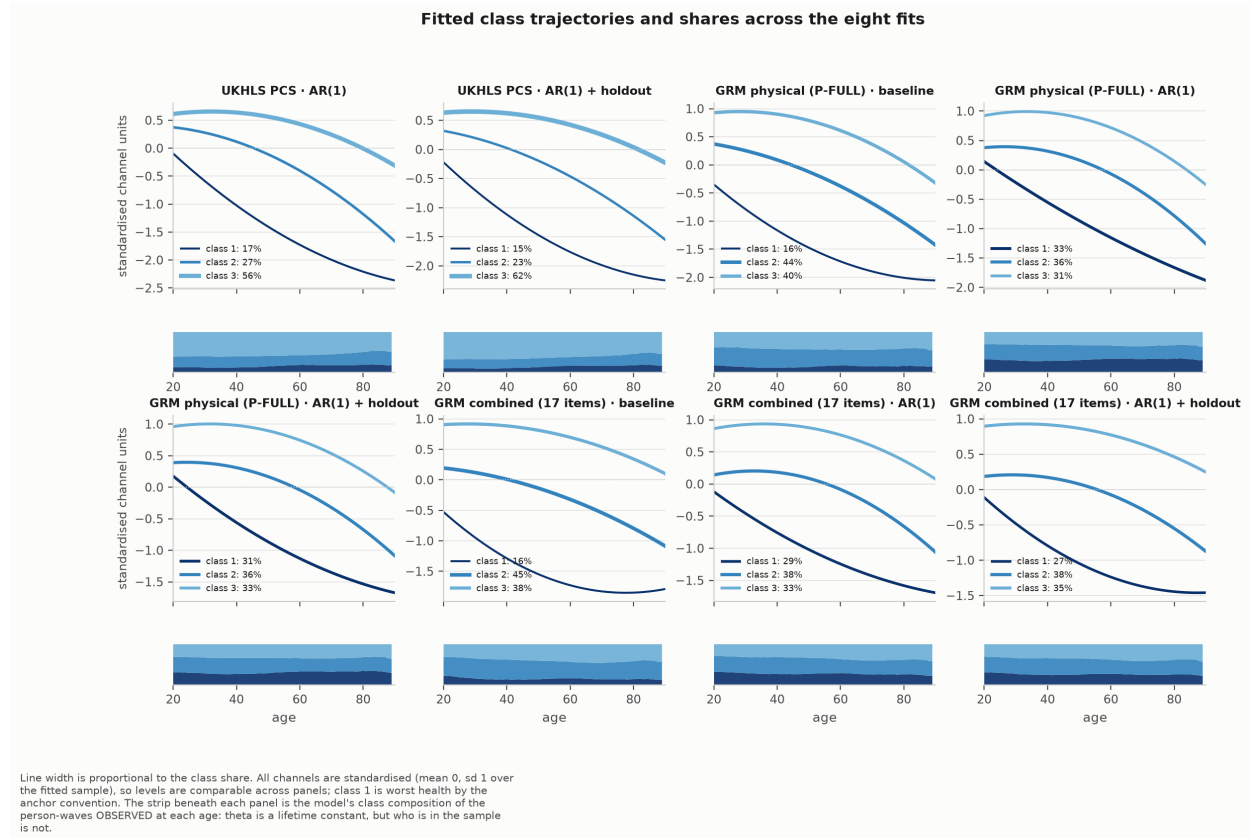


Figure 8: Fitted class mean trajectories $\alpha_k + \beta_k a + \gamma_k a^2$ (with $a = (\text{age} - 55)/10$) for all eight fits, on each channel’s standardised scale. Channels: the UKHLS PCS, the physical GRM in its P-FULL specification (`theta.phys.full`: functioning plus diagnosis groups), and the combined seventeen-item GRM (`theta.combined`). Line width is proportional to the class share.

Figure 8 puts every fit on one standardised axis. Three regularities hold everywhere: the three classes are ordered in level at every age and never cross; all three decline; and the worst class declines *fastest*, so the classes fan apart over the life course. The gap between the best and worst class widens from 1.2–1.9 sd at age 30 to 1.8–2.2 sd at age 80.

Fit	worst-class share	level gap (best – worst) age 30	age 80	decline, age 30–80 worst	best
PCS, AR(1)	.165	1.24	2.20	–1.62	–0.66
PCS, AR(1) + holdout	.146	1.35	2.19	–1.45	–0.61
GRM phys, baseline	.159	1.74	2.06	–1.22	–0.89
GRM phys, AR(1)	.328	1.21	1.81	–1.44	–0.84
GRM phys, AR(1) + holdout	.313	1.22	1.79	–1.32	–0.75
GRM comb, baseline	.163	1.86	2.19	–0.90	–0.57
GRM comb, AR(1)	.290	1.38	1.95	–1.12	–0.56
GRM comb, AR(1) + holdout	.271	1.41	1.91	–0.96	–0.46

Standardised channel units. Class 1 is worst health.

The specifications differ in *how* they separate the classes, and the difference is systematic. Adding the AR term to a GRM channel moves the worst class up in level (its age-30 gap to the best class narrows from 1.74 to 1.21 on the physical channel) while making it decline more steeply ($-1.22 \rightarrow -1.44$) and roughly doubling its membership. In other words the baseline finds a *small, extreme, comparatively flat* worst class; with persistence available the model instead finds a *larger, less extreme, faster-declining* one. The ratio of worst-class to best-class decline — a compact summary of how much of the story is trajectory divergence rather than fixed level differences — rises from 1.37 to 1.72 (physical) and 1.57 to 2.00 (combined) when persistence is added, while the PCS sits at 2.4–2.5 in both its specifications.

The holdout fits track their full-sample siblings closely on every one of these quantities (shares within .02, level gaps within .11 sd, declines within .18 sd), so the restriction to people with five or more observations changes the estimated typology very little.

Out-of-sample prediction

What the model’s held-out density conditions on. As the eight fits were run, the Stan generated quantities restarted the AR(1) recursion at the first held-out row with the *stationary* variance, so the person’s own last fitted value was not carried forward through ρ . The fitted rows reach the held-out rows only through the class posterior (the second held-out row does condition on the first). That is a coherent estimand — what the latent class structure alone predicts — but it is not the forecast a persistence model is capable of, and it understates what these fits know. The model has since been corrected (§9); the results below were computed offline on the same rows, with the offline machinery first validated by reproducing the fits’ own stored held-out densities to a correlation of 0.9999998. The AR-conditional forecast uses the same class posterior but carries the last fitted deviation forward, $\mu_k(a_h) + \rho_k^G(y_{\text{last}} - \mu_k(a_{\text{last}}))$, with the gap-adjusted predictive variance.

Two benchmarks are estimated on the fitted rows only: an age-quadratic Gaussian, and last-observation-carried-forward (LOCF), which is a strong naive predictor for a persistent quantity like health.

Channel	Predictor	LPD/obs	RMSE	R^2	cover ₅₀	cover ₉₀
PCS	model, AR-conditional	−1.005	0.671	.614	.56	.89
	model, class-only (Stan)	−1.043	0.706	.573	.60	.90
	LOCF	−1.118	0.740	.531	.61	.90
	age-quadratic	−1.416	0.991	.159	.52	.88
GRM phys	model, AR-conditional	−0.889	0.589	.672	.54	.91
	model, class-only (Stan)	−1.015	0.665	.582	.53	.91
	LOCF	−0.994	0.653	.596	.56	.90
	age-quadratic	−1.376	0.958	.133	.50	.90
GRM comb	model, AR-conditional	−0.961	0.635	.627	.54	.90
	model, class-only (Stan)	−1.079	0.709	.535	.54	.90
	LOCF	−1.058	0.697	.551	.56	.90
	age-quadratic	−1.452	1.030	.020	.48	.89

Per held-out observation (two per person; 58–79k rows), on the standardised channel scale. 400 posterior draws. cover₅₀ and cover₉₀ are the empirical coverage of central predictive intervals with nominal 0.50 and 0.90.

Age alone predicts almost nothing at the person level. The age-quadratic benchmark reaches $R^2 = .02$ –.16. Everything interesting in these data is between-person, which is the premise of the whole exercise and is worth having measured.

The class structure alone does not beat carrying the last observation forward. On the model’s own estimand, the mixture edges LOCF on the PCS (R^2 .573 vs .531) but *loses* to it on both GRM channels (.582 vs .596; .535 vs .551). A three-class typology plus an age polynomial is roughly as informative about a person’s next two observations as simply assuming they will be where they were last time — no better on the sharper measures. Stated plainly: the classes describe lifecycle shape well, but they are a coarse summary of an individual’s current position.

Persistence is what makes the model predict. Carrying the last deviation forward lifts R^2 to .61–.67, clearly ahead of LOCF on every channel, and improves the log density by .04–.13 nats per observation. The gain over LOCF is largest on the physical GRM (.672 vs .596). The two ingredients are complementary: the class trajectory supplies where the person is heading, ρ supplies how much of where they are now to carry with them, and LOCF is the degenerate case that keeps the level and ignores the trajectory.

The physical GRM predicts best on every metric (LPD −0.889, RMSE 0.589, R^2 .672), ahead of the combined bank (.627) and the PCS (.614). This is the floor analysis (§7) showing up in out-of-sample terms: a measure that resolves the ill tail finely gives the model more to predict *with* as well as more to predict.

Calibration is good at the tails, mildly conservative in the middle. 90% intervals cover 88–91%, close to nominal. 50% intervals cover 48–61%, over-covering on most fits — and the same pattern appears for the naive benchmarks, so it reflects health changes being more sharply peaked than Gaussian rather than a defect of the mixture. A heavier-tailed observation model (Student- t residuals) is the natural refinement.

The corrected generated quantities

The measurement above is a defect worth fixing rather than documenting, so `person_class_loglik` now takes a `prev_row` argument that anchors the AR(1) recursion: 0 opens a window at the stationary variance (correct for a person’s first fitted row), a row index opens it by carrying that row’s deviation

forward. The generated-quantities block emits both estimands rather than replacing one with the other — `log_lik_heldout` is now AR-conditional, and `log_lik_heldout_marginal` preserves the class-only quantity these eight fits reported.

The model block is untouched: it passes `prev_row = 0`, and `log_prob` at a fixed parameter point returns bit-identical values before and after the change (-15686.1620000000 on the test payload). The existing posteriors therefore remain valid — only the reported held-out quantity changes. `tests/test_gq_heldout.py` recomputes both estimands in numpy from the same payload and draws and requires agreement to 10^{-5} , and checks that they coincide exactly when `ar_mode == 0`.

What the corrected block reports on these fits, per held-out observation (the joint density of the two-row window, so the second row conditions on the first’s realised value — which is why it exceeds the row-by-row forecast in the table above):

Channel	class-only (as run)	AR-conditional (corrected)	gain
PCS	−1.043	−0.964	+0.079
GRM physical	−1.015	−0.836	+0.180
GRM combined	−1.079	−0.908	+0.171

So the correction is worth 0.08–0.18 nats per held-out observation on the reported density, and most on the physical GRM — the channel whose persistence estimate is the strongest. Held-out numbers from the eight fits in this note are the class-only column, and should be read as class-structure diagnostics; anything refitted from here reports the corrected quantity.

Caveats. The holdout samples are people with at least five observations, who are healthier and better-retained than the full roster; the holdout fits reproduce their full-sample siblings’ structure closely (θ within .02, ρ within .02 on every channel), which is reassuring but does not make them representative. And the combined-channel fits inherit §6’s unidimensionality caveat: a single class structure over two correlated dimensions.

9.1 How much of that convexity is real?

The figure above plots utilisation against *percentiles* of each measure, and the claim of convexity deserves more scrutiny than a percentile axis can give it. Two separate things inflate it.

The horizontal scale. Percentiles are a rank transform, so equal horizontal steps are equal shares of people rather than equal amounts of health. For a roughly normal θ the bottom decile spans far more of the health scale than the fifth does, and compressing it to the same width bends the left-hand end upward. Panels (a) and (b) of Figure 9 show the same relationship on the two axes: the sharp elbow on the percentile axis becomes a smooth decay on θ . The binning rule itself turns out not to matter — equal-count and equal-width bins trace the same curve — so this is the axis, not the aggregation. The formal test in the table above was already run on the measure’s own scale, so only the picture was affected; but the picture is what carried the claim.

The vertical scale, which matters more. In-patient stays run at 8.2%, and the logistic curve is convex everywhere below one half. A relationship that is exactly *linear in the log-odds* therefore appears convex when plotted as a probability. Panel (b) draws a logistic with no quadratic term at all against the observed decile means: it reproduces the curvature almost exactly. Panel (c) puts the same data on the log-odds scale, where it is close to a straight line.

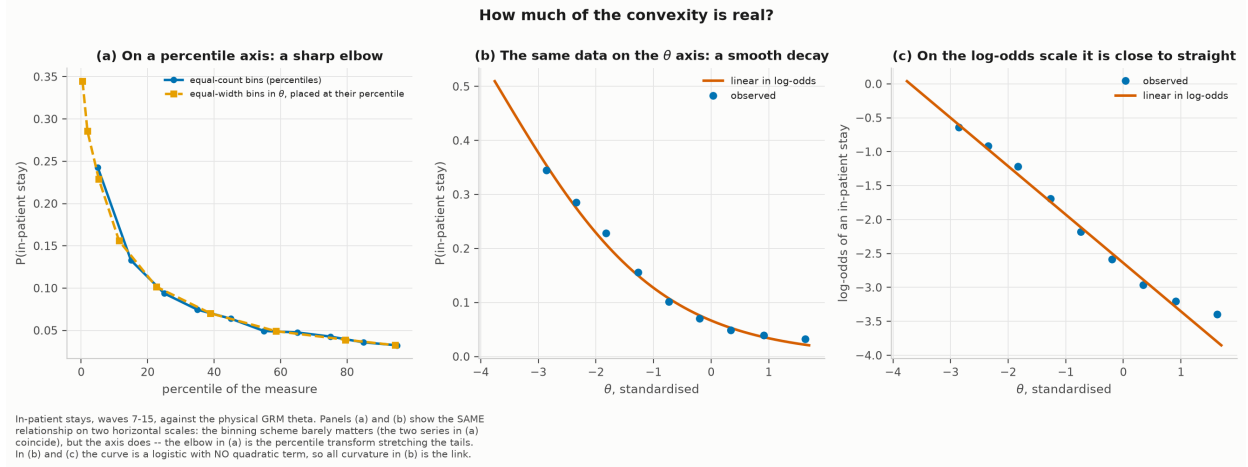


Figure 9: In-patient stays against the physical GRM θ . The fitted curve in (b) and (c) is a logistic with no quadratic term, so every departure from a straight line in (b) comes from the link function alone.

Re-running the quadratic test on both scales makes the point numerically:

Measure	probability scale		log-odds scale	
	z^2 coef	t	z^2 coef	t
GRM physical, θ	+.020	44.7	+.051	8.6
SF-6D utility	+.021	39.8	+.112	18.1
UKHLS PCS	+.011	23.7	–.028	–4.8
GRM physical, TCC scale	+.009	17.4	–.056	–10.2

On the probability scale every measure is emphatically convex; on the log-odds scale two of the four change sign. The PCS and the TCC cardinalisation are mildly *concave* in log-odds. Only θ and the SF-6D remain convex there, and θ 's curvature is small against its linear slope of -0.71 per standard deviation.

What this means for the framework. The framework asks whether the return to preventing a unit of health loss is larger for the already-unwell, which is a question about the second derivative of a structural relationship — and a second derivative is not invariant to how either axis is cardinalised. Three choices have to be made explicit before the question has an answer: the health scale (θ , its TCC transform, and the PCS rank people almost identically but disagree about curvature), the outcome scale (probability, log-odds, or eventually cost), and, if cost is the target, the fact that we observe utilisation counts rather than spend at all. The honest summary of §8 is therefore narrower than it first appeared: *utilisation falls steeply with health among the unwell and flattens among the healthy, and on the natural multiplicative scale for a risk that relationship is close to log-linear.* A convexity claim strong enough to carry a policy conclusion needs a cost outcome and a defended cardinalisation, neither of which UKHLS supplies on its own.

9.2 What the literature actually establishes

Convexity of healthcare costs in health is widely assumed in prevention arguments, so it is worth separating what is *accepted*, what is *measured*, and what is actually *known* about the functional

form. The three come apart.

Firmly measured, and about the distribution rather than the shape. Spending is extremely concentrated: the top decile of spenders accounts for more than half of total expenditure, and about a third of them are still in the top decile a year later. This is robust and uncontroversial, but it is a fact about the *marginal distribution of spending*, not about the derivative of cost with respect to health. A concentrated outcome is consistent with many shapes, including a linear one applied to a skewed health distribution.

Firmly measured, and genuinely about shape — but on the count scale. The multimorbidity literature reports a “near-exponential” relationship between the number of chronic conditions and cost, with expenditure roughly doubling per additional condition and specific combinations costing more than the sum of their parts (superadditivity). Here the scale is defensible in a way ours is not: “one more condition” has a meaning that “one more standard deviation of θ ” does not. But note what the finding *is*: exponential in the count is exactly linear in log cost. The headline result of that literature is a multiplicative model, which is the same shape §9.1 found in our data — not evidence of curvature beyond multiplicative.

Assumed by the standard specification, and therefore weakly tested. The default model for healthcare costs in health economics is a GLM with a log link (Manning & Mullahy 2001; Jones 2010), under which covariates act multiplicatively by construction. A log-link model is convex in levels automatically, whatever the data say. So a large share of the empirical literature cannot be evidence *for* convexity in levels, because it imposes it; the interesting question — is there curvature on the log scale? — is rarely posed. We found none of the reviewed work testing convexity against a continuous health measure as a functional-form question.

A rival explanation for the shape. The “red herring” literature (Zweifel, Felder & Meiers 1999, and the twenty-year reappraisals) argues that costs track *proximity to death* rather than age or health level, with spending tripling in the final quarters of life. If most of the steepness at the bottom of the health distribution is people close to death, then the apparent convexity in health is partly a survival gradient wearing a health costume — which our data cannot separate, because §11 left mortality out of the fitted models.

And a caution about the policy inference. Even granting convexity, Rose’s prevention paradox (1981, 1985) blocks the automatic move from “the return per person is highest at the bottom” to “target the bottom”: most cases arise from the large moderate-risk middle rather than the small extreme tail, so aggregate returns can favour population-wide action even when individual returns are convex. Convexity in the individual return function and optimality of targeting are different claims.

Where this leaves us. The proposition that costs rise steeply as health deteriorates is well measured and not in doubt. The stronger proposition the framework needs — that the second derivative is positive on a defended scale, so that a unit of prevention is worth more to the already unwell — is much less established than its ubiquity suggests: it is usually assumed through a log link, demonstrated on a count scale where it amounts to log-linearity, or inferred from concentration statistics that do not identify a shape. Our own finding of near-log-linearity (§9.1) sits comfortably within the literature rather than against it. Testing the framework’s version of the claim needs a cost outcome rather than utilisation counts, a health scale defended on grounds other than convenience, and mortality in the model so that proximity to death is not doing the work.

10 Towards a multidimensional model

The predecessor’s five-channel mixed model used PCS and MCS (Gaussian), self-rated health (ordered logit), a chronic count (negative binomial) and an ADL count (hurdle negative binomial). Everything in this note argues for a different set, and the binding constraint is coverage.

Candidate channel	person-waves	persons	limiting gate
θ physical, P-FUNC	497,423	84,679	none (every wave)
θ mental, without DEPR	489,278	81,729	none (every wave)
θ mental, as fitted (with DEPR)	379,462	67,017	diagnosis inventory
θ physical, P-FULL	387,599	69,958	diagnosis inventory
chronic count	76% of person-waves		diagnosis inventory
ADL count	4 waves ($\sim 8\%$)		scheduled module

One weak item is costing a quarter of the sample. The mental bank as fitted includes ever-diagnosed depression, which inherits the BHPS exclusion of §3.4. Dropping that single item — one hurdle, $a = 0.93$, the weakest item in the bank bar the rare diagnoses — recovers 109,816 person-waves and 14,712 people, a 22.4% increase. Jointly observing a physical and a mental channel then covers 484,832 person-waves on 81,458 people (49,545 of them with three or more observations), against 376,019 / 66,841 / 38,478 if either channel carries a diagnosis item. That is a 29% larger estimation sample for the loss of almost no information.

Recommended specification. Three channels, and the ragged-channel machinery already in `runner/mixed.py` handles them as they stand:

1. **θ physical (P-FUNC)**, Gaussian. Replaces PCS and, because self-rated health enters as the GH item and functioning as FUNC, it also absorbs the old model’s separate self-rated-health channel and its ADL channel — the latter traded for `disdif`, which measures the same construct in every wave rather than four.
2. **θ mental without DEPR**, Gaussian. Replaces MCS, without the orthogonality artifact, and with GHQ-12 doing most of the work.
3. **Chronic condition count**, negative binomial, on the 76% of person-waves where the inventory exists. Kept as a channel rather than folded into a measurement model, which is where §5 concluded it belongs: “ever” is the correct state variable for a diagnosis, its accumulation is a process to be modelled rather than an artefact to be corrected, and as a channel its missingness is explicit instead of silently deleting people.

Two properties of this design matter. First, the class structure is identified off channels 1 and 2, which are observed for everyone in every wave, so the BHPS members contribute to the typology even though channel 3 is missing for them — the opposite of the current arrangement, where they are dropped outright. Second, the missingness in channel 3 is a survey-design gate, not a health-related process: the excluded people are indistinguishable on every health measure (§3.4), so treating it as ignorable conditional on the observed channels is defensible in a way that ADL’s scheduled-module missingness also is, but that attrition would not be.

What to leave out, and why. Utilisation (§8) is an outcome for validating the measures, not a health dimension to be clustered on. SF-6D is a mixed physical–mental utility (§6) and would duplicate both channels at once. The combined GRM is physically dominated (Figure 5) and its

residuals say the data want two dimensions, not one. SWEMWBS, loneliness and the ADL module all fail the every-wave rule.

An extension worth considering later. Channels 1 and 2 currently compress each domain to a point estimate and then treat it as data, discarding the posterior sd the GRM already computes (§7: 0.27 near the population mean, rising to 0.65 at the extremes). Passing that through as a known measurement-error variance per observation is a small change to the Gaussian channels and would stop the model reading heteroskedastic measurement noise as heterogeneity in health — which, given the ceiling and floor results, is concentrated exactly in the classes the model cares most about.

11 The multidimensional model

The specification of §10 was fitted in four variants — baseline, holdout, AR(1), AR(1)+holdout — on one common sample of 38,268 people and 388,774 person-waves, so the four differ by specification alone. Three channels: the physical GRM (P-FUNC) and the mental GRM without DEPR, both Gaussian, and the chronic count as a negative binomial, masked on the 24% of rows with no diagnosis inventory. Mortality was toggled off for this round. All four converged.

Variant	$\max \hat{R}$	θ (shares)	ρ	chronic at 55	wall
baseline	1.0032	.189 / .377 / .434	—	2.6 / 1.4 / 0.13	8.0 h
holdout	1.0049	.187 / .371 / .442	—	2.6 / 1.4 / 0.14	6.8 h
AR(1)	1.0014	.190 / .398 / .413	.73 / .61 / .60	2.9 / 1.1 / 0.01	19.0 h
AR(1) + holdout	1.0023	.195 / .389 / .416	.72 / .58 / .58	2.9 / 1.1 / 0.01	18.2 h

“chronic at 55” is the implied mean condition count at age 55 for each class.

Extra channels stabilise the classes against the AR term. In the univariate fits of §9 adding persistence roughly doubled the worst class (.16 $\rightarrow$.33 on both GRM channels). Here the shares are flat across all four variants at about .19 / .38 / .42. The mechanism in §9 was that a single channel cannot separate a small extreme class from the autocorrelation of repeated low observations, so once ρ can absorb the latter the boundary moves; with three channels the class is identified from agreement *across* channels, and persistence no longer competes for the same variation.

Persistence is much less class-graded than the univariate fits implied. $\rho = .73/.61/.60$ here, against .78/.27/.39 for the physical GRM alone. The worst class remains the most persistent, but the other two are far more so once a second correlated channel is observed: a shock that looks transient in one measure reads as persistent when two move together. That is a warning about reading ρ off a single channel.

The classes absorb the overdispersion in the chronic count. The implied means at 55 — 2.6/1.4/0.13 conditions — weight up to 1.08 against an observed mean of 1.045, so the mixture reproduces the marginal. And ϕ_{chronic} settles at 656 (baseline) to 920 (AR), which for a negative binomial is effectively Poisson: after conditioning on class and age there is almost no overdispersion left to model.

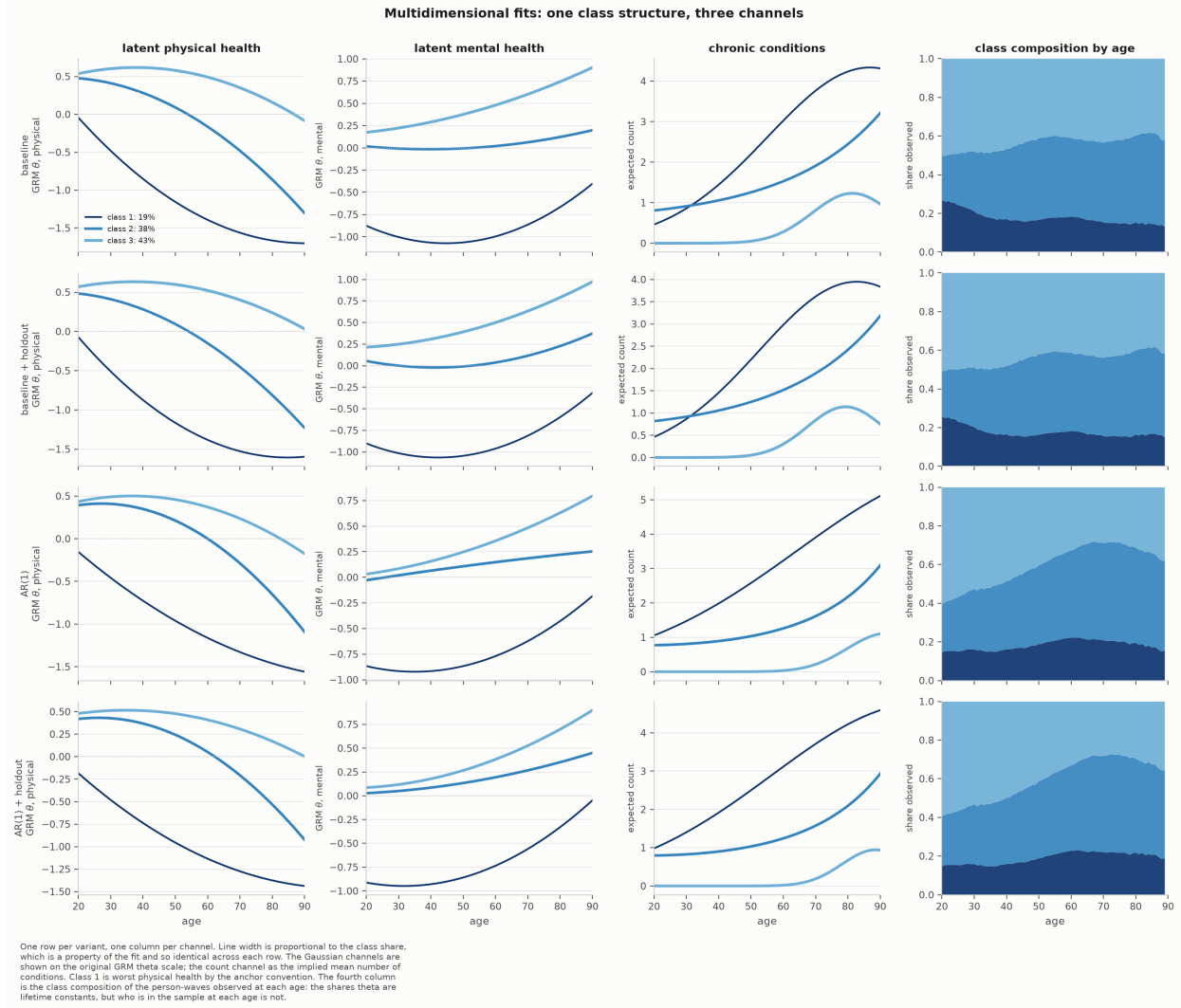


Figure 10: Class trajectories and shares for the four multidimensional fits, one row per variant and one column per channel. A fit has a single class structure, so the shares are identical across a row and the line width (proportional to the share) repeats; what changes across the row is what each class *does* on each channel. The Gaussian channels are returned to the original GRM θ scale using each fit's standardising moments; the count channel shows the implied mean number of conditions. Class 1 is worst physical health by the anchor convention. Together with Figure 8, which covers the eight univariate fits, every Bayesian model in this note has its trajectories and shares drawn.

Figure 10 makes the classes legible as types rather than as parameter vectors. The physical channel behaves as the univariate fits led us to expect: three ordered, non-crossing, accelerating declines, with class 1 falling from about -0.1 at age 20 to -1.7 at 90 while class 3 stays near $+0.5$ until the sixties. The mental channel is where the multidimensional model earns its keep. Its trajectories *rise* with age for every class, recovering the late-life wellbeing pattern of Figure 2 inside a model that is simultaneously fitting a steep physical decline — and they rise fastest for class 3, so the healthiest physical type is also pulling away mentally. Class 1 is worst on both channels at every age, which is worth stating because nothing in the model forces it: only the physical channel carries the ordering constraint, so the mental ordering is an estimate. The count channel then separates

the classes most sharply of the three, from about 4.5 conditions at age 80 in class 1 to 0.7 in class 3. The composition strips answer a question the shares alone cannot. In a mixture θ is a lifetime constant, so any movement in a strip is selection into the observed sample, not people changing class. Across the twelve fits that movement is modest but consistent in sign for the physical measures: the worst class rises from 12.1% of person-waves at age 30 to 18.8% at 80 on the PCS, and from 29.8% to 34.5% on P-FULL under AR(1) — the healthiest are observed at every age while the least healthy accumulate in the older cells. The multidimensional baselines tilt the other way (21.0% at 30 falling to 15.3% at 80), which is mortality and attrition removing the sickest, and a reminder that these strips describe who is present rather than how anyone’s health evolved.

Held-out prediction, channel by channel

Chronic is nearly free to predict — carrying the last observed count forward is exactly right on 93.4% of the 58,600 held-out chronic rows — so a single held-out number would be flattered by it. The density is therefore decomposed by channel offline, validated by reproducing the fits’ own stored per-person values (correlation 0.999999, maximum deviation 0.04).

Channel	AR(1) + holdout		holdout (no AR)
	AR-conditional	class-only	
θ physical	−0.936	−1.059	−1.121
θ mental	−1.138	−1.245	−1.321
chronic count	−0.981	−0.981	−1.086
all three	−1.021	−1.105	−1.184

Log predictive density per held-out row. 76,536 rows on each Gaussian channel, 58,623 on the count.

Three readings. First, the two Gaussian channels gain 0.12 and 0.11 nats per row from carrying the last value forward, while **the chronic channel gains exactly zero** — to three decimal places, because the model carries no persistence term on that channel at all. It predicts the count from class and age, never from the person’s own previous count.

Second, that omission is expensive. The distribution of the change from a person’s last observed count is 93.4% zero, 5.8% +1, 0.6% +2 or more, with an entropy of 0.274 nats. A channel that used the previous count could therefore reach about −0.27 per row where the fitted model achieves −0.98: roughly **0.7 nats per row left on the table**, far more than the 0.12 the AR term buys on the Gaussian side. This is the count-channel analogue of the generated-quantities defect of §9, and the obvious next fix: the chronic channel should condition on the last observed count, either as an offset or through a monotone accumulation process.

Third, the physical channel predicts better than the mental one on every variant (−0.94 against −1.14), the same ordering the univariate fits found, and the non-AR holdout has AR-conditional and class-only densities identical to four decimal places — correct by construction when there is no persistence to carry, and a useful check that the toggle behaves.

Caveats. Mortality is absent from this round, so nothing here speaks to how the classes differ in survival. The GRM thetas enter as point estimates: the posterior standard deviation the measurement model already computes (0.27 near the population mean, 0.65 at the extremes, §7) is discarded, so the mixture reads heteroskedastic measurement noise as heterogeneity in health, and does so most where the extreme classes live. Passing that through as a known per-observation

variance is a small change to the Gaussian channels and the natural next extension. Finally, the univariate comparison in §9 used P-FULL while this model uses P-FUNC, so the two are not a nested pair; a univariate P-FUNC fit would close that gap.

12 Predicting outcomes the models never saw

Everything so far scores a model against the health variables it was fitted to. This section asks a different question: given a person’s class posterior, how well can we predict outcomes that appear in *no* channel of *any* fitted model — death, and healthcare use? Mortality was toggled off for the multidimensional round and utilisation was never a channel, so this is criterion validity, and it is the comparison that decides whether the typology is useful beyond describing its own inputs.

Three outcomes: death during the panel (person-level), any in-patient stay in the last twelve months, and the out-patient attendance band 0–4 (both person-wave, waves 7–15). Three predictor sets, each also carrying a quadratic in age so that what is measured is what each adds to age alone: age only; age plus the two free class posteriors; and age plus the model’s own continuous health measure, which is the fair competitor because it is what the classes summarise. People are split 70/30, the outcome model fitted on the 70 and scored on the 30.

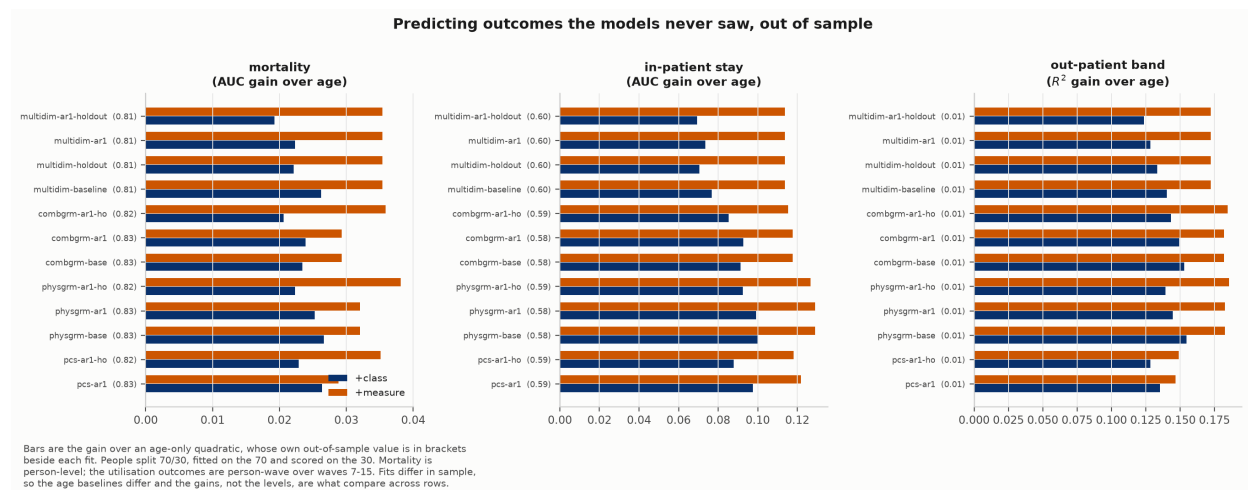


Figure 11: Out-of-sample gain over an age-only quadratic, whose own value is in brackets beside each fit. Fits differ in sample, so the age baselines differ and it is the gains, not the levels, that compare across rows.

	mortality AUC, in / out	in-patient AUC, in / out	out-patient R^2 , in / out
<i>age only</i>			
PCS sample	.836 / .831	.576 / .588	.011 / .012
multidim sample	.828 / .815	.586 / .596	.010 / .012
<i>age + class posterior</i>			
PCS, AR(1)	.856 / .857	.678 / .686	.143 / .148
P-FULL, baseline	.853 / .861	.674 / .681	.155 / .166
P-FULL, AR(1)	.852 / .860	.675 / .680	.148 / .157
combined, baseline	.850 / .856	.664 / .672	.157 / .165
multidim, baseline	.843 / .841	.667 / .673	.152 / .153
multidim, AR(1)	.841 / .837	.663 / .670	.144 / .141
<i>age + the continuous measure</i>			
P-FULL	.860 / .866	.705 / .710	.187 / .194
multidim (P-FUNC)	.855 / .850	.705 / .710	.182 / .185

The continuous measure beats the classes on every outcome, in every fit, in and out of sample. For out-patient attendance the gap is large: R^2 of .194 against .166, so discretising into three classes throws away about a sixth of what the measure knows. For in-patient stays it is .710 against .681 in AUC. This is the sharpest statement of a theme running through the note — the classes are a summary, and summarising costs. If prediction is the goal, use θ ; the typology earns its place by being interpretable and by describing trajectories, not by predicting better than the thing it summarises.

Age already predicts mortality well, and health adds little. The age-only AUC is .83, and the best class specification reaches .861 — a gain of about .027. Health measured in mid-life is a weak marginal predictor of death once age is known, at least at this horizon. Utilisation is the opposite: age alone gives AUC .59 for in-patient stays and $R^2 = .01$ for out-patient bands, and health does the overwhelming majority of the work. Prevention targeted on these measures should expect to move utilisation far more readily than mortality.

The multidimensional classes are not better than the univariate ones at this, and are slightly worse. On gain over age they are comparable for mortality (+.026 against +.027 for P-FULL baseline) but behind on both utilisation outcomes (+.077 against +.100 AUC for in-patient; +.141 against +.154 R^2 for out-patient). Adding channels bought stability and interpretation (§11), not external predictive power. Note the samples differ, so only the gains are comparable, and the multidim measure column is P-FUNC where the univariate is P-FULL.

The AR variants predict external outcomes slightly worse than their baselines, consistently across all three outcomes and both families. That is coherent with §9: the AR term absorbs within-person variation that would otherwise separate the classes, so the classes it leaves behind carry marginally less information about anything else.

Caveat. The class posteriors come from mixtures fitted on all people, so the 70/30 split tests the class-to-outcome relationship out of sample but not the whole pipeline; a fully clean version would refit each mixture on the training people. In-sample and out-of-sample figures are nonetheless close throughout — often the out-of-sample number is the higher of the two — so there is little sign of the outcome models overfitting.

13 Recommendations

1. Physical health for lifecycle dynamics: **P-FUNC** (or the original GRM). Mental health: **MENT**. Report both; their correlation (0.457) is a finding, not an assumption.
2. **P-FULL** as the descriptive companion when condition burden should show in the score; P-REC/P-TIMED quantify its recency sensitivity.
3. Conditions in the Bayesian mixture as separate channels; “ever” is the right state variable there.
4. Avoid the combined score unless a single index is externally required; if it is, build the bifactor version first.
5. SF-6D: treat as a mixed physical–mental utility (its closest GRM cousin is the combined bank), never as a physical measure; licence before publishing utilities.
6. Report θ and `grmh` together when consequences matter: θ is the latent-normal unit the models want, `grmh` the near-linear-in-utilisation unit (§8); the choice of cardinalisation is substantive, not cosmetic.

14 Reproducibility

<code>data_cleaning/03_build_measures.py</code>	SF-12/SF-6D layer; validates construction note (max err 0.0)
<code>data_cleaning/04_build_grm.py</code>	rows, the 6.6-pt artifact, tariff
<code>data_cleaning/05_build_chronic.py</code>	original GRM replicated to 498,424 person-waves
<code>data_cleaning/06_build_grm2.py</code>	chronic layer; 100% timing
<code>descriptives/01--12</code>	EIT count on 100.000% of rows
<code>clustering/run_fit.py, runs/overnight_queue.py</code>	all GRM-2 specs; Q3, EAP id
<code>clustering/run_multidim.py, runs/multidim_queue.py, multidim_channel_decomp.py</code>	ance, recency weights
	measure panel, correlations
	ure, the ceiling/floor, conv
	composition and outcome-pred
	the §9 batch: generic registry
	observations holdout, per-per
	densities
	the §11 batch and its per-char
	decomposition

Everything under `data/` derives from licensed UKHLS microdata and is never committed; all figures are aggregate statistics.

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